

GMU IT 357 Forensic Report

John Murray

This is an educational project for a George Mason University course. Contents of this report are part of a fictional scenario.

THIS IS AN EDUCATIONAL PROJECT

By submitting this assignment, I certify I have abided by all requirements of the GMU honor code. I certify that this is entirely my own work, no unauthorized sources have been used, and all sources used have been properly cited.

Investigator Information

My name is John Murray, and I am a computer forensic examiner working for the Federal Bureau of Investigation (FBI). I have around "12" years of forensic experience with the FBI.

Case Background

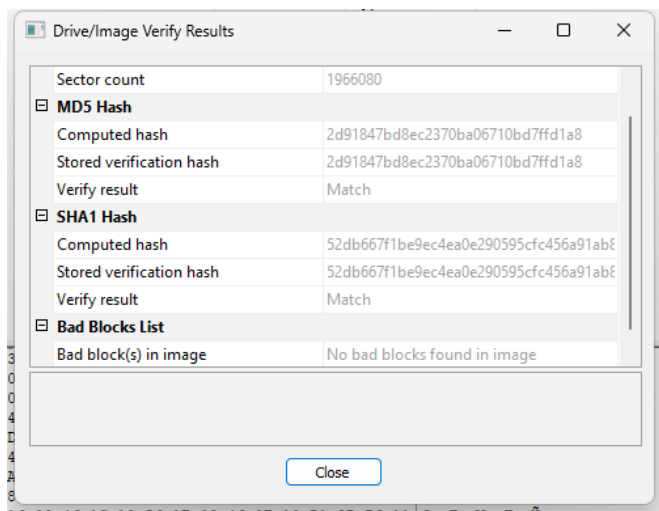
The Federal Bureau of Investigation (FBI) is investigating a series of threats that were posted on social media by James T. Buchs, who is a graduate student pursuing a Master of Arts in Political Science at George Mason University (GMU). Buchs has been residing in Beacon Hall at GMU's Science and Technology campus in Manassas, Virginia, and he has been reported twice to GMU's Violence Prevention Committee (VPC). After evaluation of his social media posts, the FBI determined that there was probable cause that indicates Buchs was a threat to the safety of the students and faculty at GMU. Thus, a search warrant was authorized for Buchs residence at 10945 George Mason Cir #212, Manassas, VA 20110. During the execution of the warrant, the FBI found a flash drive and was transported to the FBI forensic laboratory, where a forensic examiner produced a digital forensic image of the drive verified by hash values. This forensic image was then provided to me on 4/15/2025 for detailed forensic analysis.

Evidence Analyzed

The evidence I am analyzing and examining is a forensic image of a flash drive that was seized from James T. Buchs residence, created by the FBI forensic techs. The flash drive was found at Buchs' apartment at 10945 George Mason Cir #212, Manassas, VA 20110. I obtained the verified forensic image file and associated hash values directly from the FBI lab for my detailed forensic analysis.

Verification of Evidence Integrity

To verify the evidence integrity, after downloading the image file I began to verify the hash from the suspect's flash drive was matching the evidence image file received, for this process I utilized FTK Imager. On FTK imager, after adding the image file to FTK imager I right clicked on the IT-357-GMU-Buchs.E01 and used the verify drive/image selection to read the verified hashes. A hash is a cryptographic function like a mathematical algorithm that converts data into a fixed sized string of text that is unique which is the hash value and is important because it helps encrypt and decrypt digital signatures which verify the authenticity and integrity of a digital file. The MD5 hash was 2d91847bd8ec2370ba06710bd7ffd1a8, which was verified and confirmed to be the same.



Forensic Tools

The forensic tools that I am using for this are Exterro FTK Imager 4.7.3.81 and Autopsy 4.21.0.

Overview

Before starting the forensic analysis, I validated all of my hardware and software tools to ensure proper functionality of my system and the applications. I then duplicated the forensic image provided by the FBI forensic techs and verified its authenticity using hash values. Once I confirmed the validity, I used Autopsy to extract and review the data, organizing the investigation by developing a list of directories and using it to guide the analysis. Any possible evidence pieces were marked as possible evidence (further focus needed). The drive structure included both automated and manual searches that I used to analyze the forensic image.

Documents

During the investigation of this forensic image, three documents were discovered in relation and lead investigation. The documents were found after opening the forensic image in the Autopsy application and verifying the hash value. After creating a new case in Autopsy, I then created a folder to store possible forensic data extracted. After adding the forensic image into the case, autopsy then processed the data and showed a structured view of the drive with recovered contents.

Security.docx

While going through the drives directories I found a document named "Security.docx". This document was found in IT357-GMU-Buchs.E01_1 Host/IT357-GMU-Buchs.E01/Security.docx. The file was created on 2023-03-27 at 17:06:38 EDT and located in unallocated space (the area that hasn't been assigned to a partition, meaning it's not currently being used for data storage) with an internal ID of 221. The file was created using Microsoft Office Word. This file was deleted from the drive and that is known from the red "X" marked on the file. If the file was not deleted the file would not have a red "X" displayed on the file.

Appendix Item #1

Manifesto.docx

After finding the previous document, I then came across "Manifesto.docx". The file was found in IT357-GMU-Buchs.E01_1 Host/IT357-GMU-Buchs.E01/Manifesto.docx and was deleted from the original drive as seen by the red "X" on the file.

Appendix Item #2

diaz, a - risk communication and the vt massacre.doc

This file was found at /img_IT357-GMU-Buchs.E01/diaz, a - risk communication and the vt massacre.doc and modified on 2024-01-10 at 09:23:48 EST. Talking about the importance of communication during crisis.

Appendix Item #3

Graphics

The process of finding the graphics is very similar to the documents section above. When going through the directories of the image I found 3 graphics to aid in investigation.

Images (1).jpg

I found this jpg image file when going through the directories, specifically in IT357-GMU-Buchs.E01_1 Host/IT357-GMU-Buchs.E01/images (1).jpg. This file was deleted from the original drive and can be seen by the red "X" displayed in the image file name. I recovered this image by extracting the image to my forensics file with forensic evidence inside.

Appendix item #4

22-060-Fairfax-Map.pdf

During investigation I came across a map file of Fairfax, created on 1980-01-01 at 00:00:00 EST and last accessed on 2024-01-10. The file was found in IT357-GMU-Buchs.E01_1 Host/IT357-GMU-Buchs.E01/22-060-Fairfax-Map.pdf and the file was not deleted however there was a deleted version that I did not include.

Appendix item #5

university-4-rt-er-231120_1700513176407_hpEmbed_3x2.jpg

Through investigation this image file was found that was created on 2024-01-10 with an internal ID of 444. The file was found in IT357-GMU-Buchs.E01_1 Host/IT357-GMU-Buchs.E01/university-4-rt-er-231120_1700513176407_hpEmbed_3x2.jpg, in allocated space. The file was not deleted and is proved by no red "X" in the file listing.

Appendix item #6

Deleted Files

During the investigation I found multiple deleted files in relation to the investigation. After opening the image in Autopsy and ensuring the verified hash value, I proceeded to the deleted files tab under the file views expanded section under the data sources in the left side of autopsy. This section of autopsy displays the deleted files that were recovered in the creation of the forensic image.

Links.docx

The first deleted file that I came across when investigating was found in the IT357-GMU-Buchs.E01_1 Host/IT357-GMU-Buchs.E01/Deleted Files/All/Links.docx and created on 2023-03-27 at 17:06:55 EDT. This file contains 5 links with bomb making awareness and response timeline in Michigan school shooting.

Appendix Item #7

_igure2.jpg

A deleted file created on 2023-03-27 with an internal ID of 233. The file was found in IT357-GMU-Buchs.E01_1 Host/IT357-GMU-Buchs.E01/Deleted Files/All/_igure2.jpg. The file contains an image of some kind of bullets.

Appendix item #8

[_igure3.jpg](#)

A deleted file created on 2023-03-27 with an internal ID of 187. The file was found in IT357-GMU-Buchs.E01_1 Host/IT357-GMU-Buchs.E01/Deleted Files/All/_igure3.jpg. The file contains an image of an assortment of weapons.

Appendix item #9

HTML or Web-based Files

The process and procedure of recovering HTML or Web-based Files is very similar to what is described above on the documents and graphics.

remote detonator - Bombshock Forums.htm

This file is a linked file from visiting a forum on the topic of remote detonators created on 2023-03-27. The HTML was found in /img_IT357-GMU-Buchs.E01/remote detonator - Bombshock Forums.htm.

Appendix item #10

Bomb-Making Materials Awareness Program (BMAP) _ CISA.html

This file is a HTML file on Bomb making material awareness created on 2023-03-27. The HTML was found in /img_IT357-GMU-Buchs.E01/ Bomb-Making Materials Awareness Program (BMAP) _ CISA.html.

Appendix item #11

Beslan school siege Russia 'failed' in 2004 massacre - BBC News.html.

This file was found at /img_IT357-GMU-Buchs.E01/Beslan school siege_ Russia 'failed' in 2004 massacre - BBC News.html. This file contains news about a school siege in Russia that “failed”. It was created on 2023-07-27

Appendix item #12

OLE Subitems

In Autopsy, I began by creating a new case and loading the suspect's disk image to initiate the forensic investigation. I then navigated to the Data Artifacts section and selected Extracted Content, which could then reveal embedded files hidden within other files like emails or containers. These OLE Subitems are crucial because they often contain metadata, hidden

content, or embedded evidence that can provide insight into document authorship and usage history in cybercrime cases.

university-2-rt-er-231120_1700512742812_hpEmbed_3x2.jpg

This file contains an embedded image of a photo with a sign saying, “Jews for Palestine”. This file was created on 2024-01-10 at 09:12:59 EST. It was found at /img_IT357-GMU-Buchs.E01/The Israel-Hamas war has college campuses on edge. How some are tackling the issue. - ABC News_files/university-2-rt-er-231120_1700512742812_hpEmbed_3x2.jpg.

Appendix item #13

Summary

A verified forensic image of a flash drive seized from the suspect’s residence that I obtained was analyzed using FTK Imager 4.7.3.81 and Autopsy 4.21.0. The drive contained deleted and recovered files, including documents (Security.docx, Manifesto.docx, diaz, a - risk communication and the vt massacre.doc), images (Images (1).jpg, _igure2.jpg, _igure3.jpg), and HTML files referencing bomb-making forums and awareness sites. The image also included a Fairfax map PDF and graphics potentially tied to university-related tensions. OLE Subitems in embedded HTML artifacts revealed additional visual content, such as politically charged imagery, further enhancing the evidentiary context. All recovered files were located through structured directory navigation, metadata timestamps, and deleted file listings within Autopsy, ensuring a factual and methodical examination.

Appendix

Appendix Item 1:

Security Services
Our Public Safety Division houses the Security Officers Unit. It is made up of 15-20 employees. All security officers are non-sworn officers who patrol the Arlington, Fairfax, and Prince William campuses. They are the eyes and ears of the Department of Police and Public Safety and perform building lock-downs, unlocks, fire watches, personal escorts and patrols the exterior and interior of buildings. Our goal is to provide exceptional security services with professionalism, compassion, and dignity, while dedicated to excellent customer service and a safe and secure campus. All officers receive over eighty hours of training including CPR/AED certification and are certified Campus Security Officers by the Virginia Department of Criminal Justice. For security service requests, please call the Security Operations Manager, TJ Juris at 703-993-6166 or via email.

Welcome
Welcome to George Mason University Department of Police and Public Safety, we serve in one of Virginia's largest universities. George Mason University serves more than 35,000 students on its Fairfax, Arlington, Prince William and Loudoun Campuses. Situated on 677 acres of wooded land in the heart of Fairfax County, the Fairfax Campus is just 16 miles outside of Washington, D.C. Near the Virginia Square-GMU Metro, the Arlington Campus houses the Antonin Scalia Law School and emphasizes courses on law, economics and public policy. The Prince William Campus covers 124 acres in the Manassas area and specializes in bio-sciences, bioinformatics and biotechnology, as well as computer and information technology. The University Police Department patrols three of the four campuses 24 hours a day, 365 days of the year. This website introduces you to the personnel, services, proper security measures, policies and regulations that promote campus safety. You will find that the University Police Department provides many services that will assist you to safely complete a college education. Special safety tips and the directory of relevant telephone numbers are available to help guide you. Take advantage of the many programs and services that we offer to the GMU community, our programs promote safety and are available to provide assistance.

Mission Statement
The mission of the George Mason University Department of Police and Public Safety is to provide a safe and secure environment for all university employees, students, and guests. Embracing the principles of community policing, the Department will work collaboratively with academic and administrative units, individuals, and organizations in furtherance of the mission of the University.
The George Mason University Police Department is responsible for protecting constitutional guarantees and impartially enforcing the law. We believe that integrity is the basis of public trust and that honesty and equality in the delivery of police services is essential. We commit ourselves to uphold these values and to foster cooperation and respect within our community.
As the law enforcement department of a state institution of higher learning, the Department will always strive to incorporate best practices in its policies, and operate with a commitment to equality, fairness, openness, and transparency with the University community.

Contact Us
- Emergency – 911
- Non-Emergency – (703) 993-2810
- TDD/TTY – (703) 993-2817
- Reporting a Crime (Crime Solvers Anonymous Tip Hotline) – (703) 993-4111
- Physical Address – 4393 University Drive, Fairfax, VA 22030
- Mailing Address – 4400 University Drive, MS 3D3, Fairfax, VA 22030

Campus Emergencies
Browse through the emergency situations listed below to review Mason's suggested response procedures, and to access relevant emergency preparedness guides per topic. Questions? Contact safety@gmu.edu.

Bomb Threat	Type of Emergency	Suggested Response	Corresponding Guide(s)
Bomb Threat	If a bomb threat is received by phone:		
	Remain Calm, Contact or ask someone to contact University Police, 703-993-2810.		

2. The industrial-technological system may survive or it may break down. If it survives, it MAY eventually achieve a low level of physical and psychological suffering, but only after passing through a long and very painful period of adjustment and only at the cost of permanently reducing human beings and many other living organisms to engineered products and mere cogs in the social machine. Furthermore, if the system survives, the consequences will be inevitable: There is no way of reforming or modifying the system so as to prevent it from depriving people of dignity and autonomy.

3. If the system breaks down the consequences will still be very painful. But the bigger the system grows the more disastrous the results of its breakdown will be, so if it is to break down it had best break down sooner rather than later.

4. We therefore advocate a revolution against the industrial system. This revolution may or may not make use of violence; it may be sudden or it may be a relatively gradual process spanning a few decades. We can't predict any of that. But we do outline in a very general way the measures that those who hate the industrial system should take in order to prepare the way for a revolution against that form of society. This is not to be a POLITICAL revolution. Its object will be to overthrow not governments but the economic and technological basis of the present society.

5. In this article we give attention to only some of the negative developments that have grown out of the industrial-technological system. Other such developments we mention only briefly or ignore altogether. This does not mean that we regard these other developments as unimportant. For practical reasons we have to confine our discussion to areas that have received insufficient public attention or in which we have something new to say. For example, since there are well-developed environmental and wilderness movements, we have written very little about environmental degradation or the destruction of wild nature, even though we consider these to be highly important.

6. Almost everyone will agree that we live in a deeply troubled society. One of the most widespread manifestations of the craziness of our world is leftism, so a discussion of the psychology of leftism can serve as an introduction to the discussion of the problems of modern society in general.

7. But what is leftism? During the first half of the 20th century leftism could have been practically identified with socialism. Today the movement is fragmented and it is not clear who can properly be called a leftist. When we speak of leftists in this art



EDUCATION

Hex
Text
Application
File Metadata
OS Account
Data Artifacts
Analysis Results
Context
Annotations
Other Occurrences

Strings
Extracted Text
Translation

Page: 1 of - Page
Matches on page: - of - Match
100%
Reset

Topic: Risk Communication and the Virginia Tech Massacre

Description: The importance of effective communication during a crisis cannot be over-emphasized. The tragic events of April 16, 2007 on the Virginia Tech campus illustrate both the danger of ambiguity and the need for a prompt response. Many lessons have been learned from Virginia Tech's successes and mistakes. Though our false-sense of security on college campuses has been shaken, colleges around the nation are more aware of the risks and are taking steps to address them in all kinds as a result of the massacre

Presentation:

I will be discussing risk communication using the Virginia Tech (VT) shooting of April 16, 2007 as a scope. On that day Seung Hui Cho, a student at VT inspired fear and disrupted the sense of security felt by the campus community. The events that shaped this day have left their mark not only on Virginia Tech, but on most universities in the United States (US) and provide a case study for mass information dissemination, establishing Emergency Operation Centers (EOCs) on campuses, better communication between communities and counseling centers, and gun control laws.

When communicating about a crisis the content, the communicator, the tone and the means of communication all must be considered (Covello, 2001). The public hangs on the organization's every word. Transparency, continuity, control, knowledge, and resources are all vital to the crisis response. Virginia Governor Timothy Kaine acted quickly in the wake of the massacre, establishing a panel to look into the events that culminated on April 16, the incident itself, and its consequences (Davies, 2008). The panel included members of the community, sponsors, and people close to Cho in order to obtain as much information as possible. The Panel held public meetings, which not only allowed the investigators to obtain information from direct sources but also allowed the stakeholders and the Panelists allowing stakeholders' expectations to be known. The Panel noted three types of concerns in their review (Davies, 2008). The first is structural, pertaining to the federal gun laws. Ambiguity and "inconsistencies between federal and state gun laws" leave room for interpretation and those with criminal records to purchase guns this was how Cho was able to purchase two semiautomatic weapons (Davies, p. 11). Quickly following April 16, Governor Kaine made Virginia law clearer by requiring background checks for gun purchases. Other states have similarly ambiguous gun laws, and may risk a similar incident occurring (Davies, 2008).

The second concern noted by the Panel is the management of mental health services by the University and State government. More effective communication between the entities had the potential to provide better treatment from two different centers, the Cook Counseling Center located on Virginia Tech, which falls under the Family Educational Records Privacy Act (FERPA), and the Carilion St. Alban's Behavioral Health Center, which falls under the Health Insurance Portability and Accountability Act (HIPAA). The discrepancy between FERPA and HIPAA laws created a situation that contributed to Virginia Tech and the community's lack of awareness of just how serious the situation was.

Cho himself contacted the Cook Counseling Center twice (Davies, 2008). A diagnosis or treatment procedure was never made, although a counselor spoke with, and collected information from, Cho via email. Cho showed signs of extreme social anxiety and a lack of relationships, but noted no suicidal or "homicidal thoughts in counseling more than a year before the April 16, 2007" (Leinwand, 2009, p. 3A).

However, in December 2005 Cho was taken to Carilion under a Temporary Detention Order (TDO) after threatening suicide and being deemed a danger to himself or others (Davies, 2008). Carilion charges

PROJECT



LOCAL



Appendix Item 6:



Appendix Item 7:

Hex Text Application File Metadata OS Account Data Artifacts Analysis Results Context Annotations Other Occurrences

Strings Extracted Text Translation

Page: 1 of - Page < > Matches on page: - of - Match < > 100% 🔍 ⌂ Reset Text Source: File Text

<https://www.detroitnews.com/story/news/local/michigan/2023/03/16/oxford-michigan-school-shooting-sheriff-response-timeline/70012836007/>

https://dailyprogress.com/news/uva/state-investigation-of-uva-shooting-differs-from-that-of-virginia-tech/article_7ee56cbc-a826-11ed-9beb-f3ade7f262e9.html

<https://www.bbc.com/news/world-europe-39586814>

<https://si.gmu.edu/calendar/>

<https://www.cisa.gov/resources-tools/programs/bomb-making-materials-awareness-program-bmap>

-----METADATA-----

Application-Name: Microsoft Office Word
Application-Version: 16.0000
Author: Franklin B. Saunders
Character Count: 868
Character-Count-With-Spaces: 1018
Content-Type: application/vnd.openxmlformats-officedocument.wordprocessingml.document
Creation-Date: 2023-03-27T21:08:00Z
Last-Author: Pollard, Rebecca J
Last-Modified: 2023-03-27T21:09:00Z
Last-Save-Date: 2023-03-27T21:09:00Z
Line-Count: 7
Page-Count: 1
Paragraph-Count: 2
Revision-Number: 2

Appendix Item 8:



Appendix Item 9:



Appendix Item 10:

Metadata
Name: /img_IT357-GMU-Buchs.E01/remote detonator - Bombshock Forums.htm
Type: File System
MIME Type: application/octet-stream
Size: 0
File Name Allocation: Allocated
Metadata Allocation: Allocated
Modified: 2023-03-27 16:56:50 EDT
Accessed: 2023-03-27 00:00:00 EDT
Created: 2023-03-27 17:06:38 EDT
Changed: 0000-00-00 00:00:00
MD5: Not calculated
SHA-256: Not calculated
Hash Lookup Results: UNKNOWN
Internal ID: 19

From The Sleuth Kit istat Tool:
Directory Entry: 30
Allocated
File Attributes: File, Archive
Size: 0
Name: REMOTE-1.HTM

Directory Entry Times:
Written: 2023-03-27 16:56:50 (EDT)
Accessed: 2023-03-27 00:00:00 (EDT)

Appendix Item 11:

Resources & Tools
All Resources & Tools
Services
Programs
Resources
Training
Groups

Bomb-Making Materials Awareness Program (BMAP)

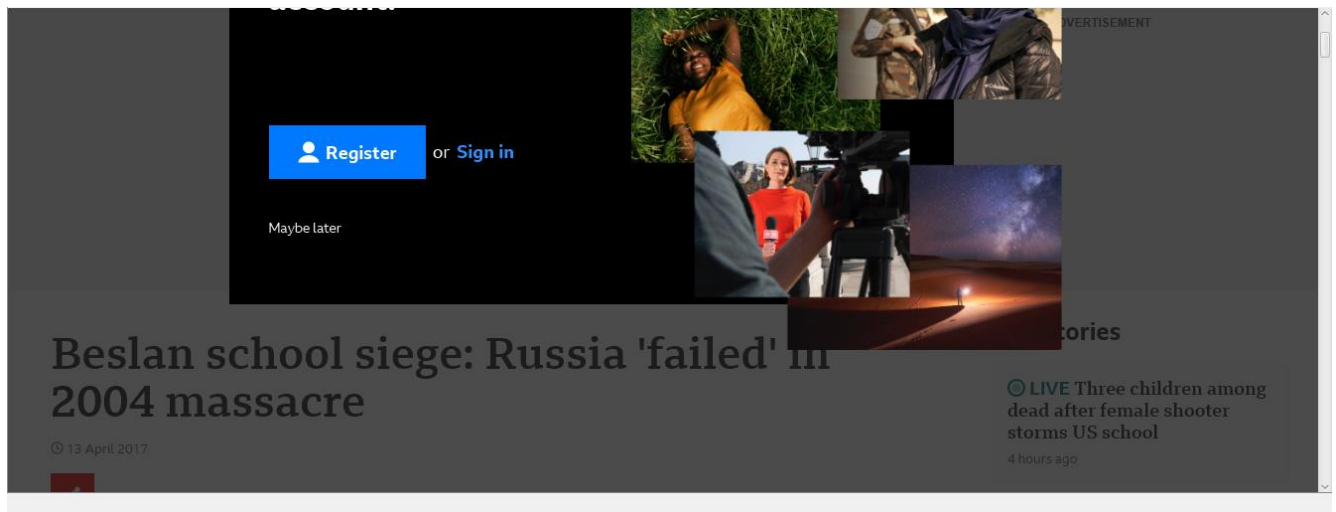
RELATED TOPICS: [PHYSICAL SECURITY](#), [BOMBING PREVENTION](#)

VISIT BOMBING PREVENTION →

Overview

The Bomb-Making Materials Awareness Program

Appendix Item 12:



Appendix Item 13:

